INCH POUND

MIL-C-50679B (AR)
24 August 1990
SUPERSEDING
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13 June 1981

MILITARY SPECIFICATION

CIRCUIT BOARD ASSEMBLY, AGC: 10549230

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an AGC Circuit Board Assembly, hereinafter referred to as an AGC Board, used in AGC Circuit Assembly (10549228). The AGC Circuit Assembly is used in Radar Receiver R-1475, part of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Main tenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization, Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10549228 - AGC Circuit Assembly
10549230 - AGC Circuit Board Assembly

SPECIAL PACKAGING INSTRUCTION

PS10549230 - Packaging of AGC Circuit Board Assembly

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10549230, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the AGC board shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The AGC board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The AGC board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The AGC board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The AGC board shall meet the requiements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The AGC board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degree F and +155 degree F.

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3.3.4.2 Non operating (storage). The AGC board shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degree F for a period of 24 hours and +160 degree F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The AGC board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the AGC board.

3.5 Performance. The assembly shall meet the requirements of 3.5.1.

3.5.1 Input and output characteristics. The AGC board shall provide two gated AGC output levels proportional to the signal strength of the video input pulse. Gating of the output signals shall be initiated by an external trigger pulse. When supplied with the inputs specified in Table I (Input and output characteristics), with the exception that the video input signal amplitude shall be 0.0 V in lieu of 1.5 V, and when adjusted in accordance with 3.5.2 thru 3.5.4, the AGC circuit board assembly shall provide the outputs specified in Table I.

TABLE I. Input and output characteristics.

REQUIREMENT	CHARACTERISTIC	TERMINALS
Input voltage	+15.0 ± 0.1 Vdc	E1, E3 (+15 V) E4, E7 (+15 V RET)
Input voltage	-15.0 ± 0.1 Vdc	E2 (-15 V) E4 (-15 V RET)
Input signal, video	Positive pulse of 1.5 V, 0.2 us wide, delayed 3.0 us from leading edge of pre-range gate trigger.	E14
Input signal, prerange gate trigger	Positive pulse of 2.5 V, 10.0 us wide with a frequency of 20.0 KHz.	E13

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TABLE I. Input and output characteristics (continued).

REQUIREMENT	CHARACTERISTIC	TERMINALS
Output to pre IF amplifier	A negative pulse having a high level voltage of 5.0 ± 1.5 V and a low level voltage of 0.0 ± 0.2 V when terminated to a resistance of 850 ohms. Pulse shall be 5.5 ± 2.0 us wide (50% points), and shall have a leading edge fall time of 0.5 ± 0.3 us and a trailing edge rise time of 0.8 ± 0.4 us (10%-90% points).	E6
Output to IF amplifier	A negative pulse having a high level voltage of 5.0 ± 1.5 V and a low level voltage of 0.0 ± 0.2 V, when terminated to a resistance of 850 ohms. Pulse shall be 5.5 ± 2.0 us wide (50% points), and shall have a leading edge fall time of 0.5 ± 0.3 us and a trailing edge rise time of 0.8 ± 0.4 us (10%-90% points).	E15

3.5.2 AR1 zero. Set AR1 zero by adjusting R60 fully counter-clockwise.

3.5.3 AR2 gain. It shall be possible to adjust the gain of AR2 using R61. With R62 adjusted fully clockwise and with an input of 1.00 Vdc to AR2, the output of AR3 shall be adjusted by R62 to 1.00 ± 0.03 Vdc.

3.5.4 AR4 gain. It shall be possible to adjust the gain of AR4 using R64. With R63 adjusted fully counterclockwise and with an input of 1.00 Vdc to AR4, the output of AR4 shall be adjusted by R64 to 1.00 ± 0.03 Vdc.

3.5.5 AGC tracking. It shall be possible to adjust the tracking of the pre-IF output relative to the IF output. With the inputs specified by Table I applied to the AGC circuit board assembly, adjust the amplitude of the input signal to E14 to obtain a low level voltage of 4.0 ± 0.1 V on E15. The low level voltage at E6 shall be adjusted to 0.6 ± 0.1 V by R62. When the input signal is further adjusted to obtain a low level voltage of 6.0 ± 0.1 V on E15, the low level voltage on E6 shall be 2.6 ± 0.2 V.

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3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practive to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 degree to +90 degree F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Circuit Board Assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II (Functional tests) and Table III (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One AGC board out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE IV. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	0 to 20 Vdc, regulated; AC ripple less than 50 millivolt (mV) peak-to-peak, 500 milliamp (mA) current.
2.	Power supply	Same as item 1.
3.	Pulse generator	Adjustable positive pulse amplitude from 0.75 to 5 V. Pulse width adjustable from 0.2 to 20 microseconds (us). Rise and fall time not greater than 10 nanoseconds (ns) PRF, 20 KHz. External triggering with adjustable delay of up to 3.0 us.
4.	Pulse generator	Same as item 3, however external triggering is not required.
5.	DC digital voltmeter	Visual display 4 digits minimum, 99.99 V, 9.999 V and 0.999 V full scale. Accuracy $\pm$0.1% of reading, $\pm$1 digit.
6.	Oscilloscope	Dual channel. Vertical rise and fall time not greater than 10 ns. Sweep lengths of 1.0 us or greater per division. Scope probe, 10 megohm input resistance.
7.	Resistive termination	850 ohm, coaxial, type BNC.
8.	Resistive termination	850 ohm, coaxial, type BNC.
9.	Test harness	Figure 1.

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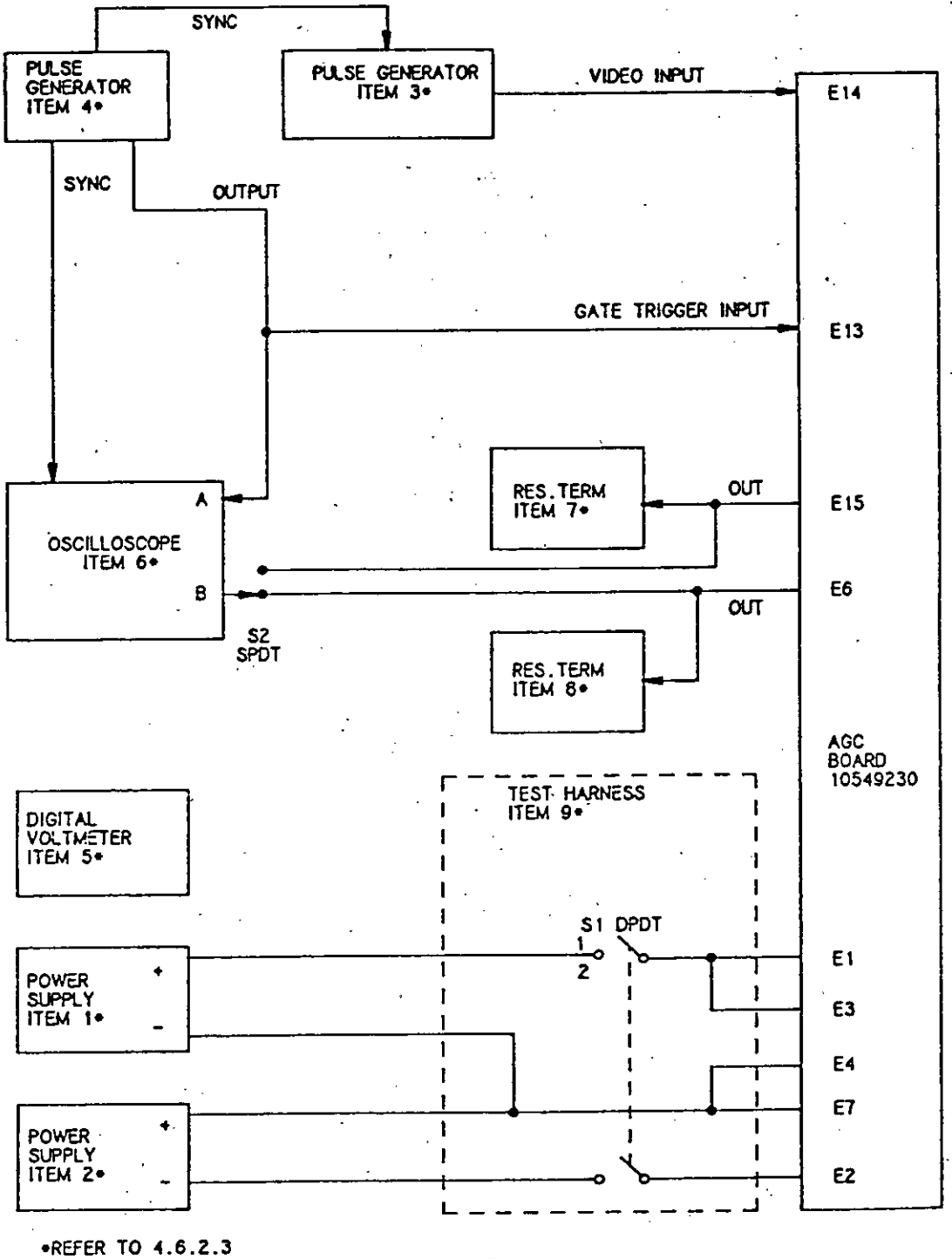


FIGURE 1. AGC board test setup.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The AGC board shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the AGC board shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the AGC board shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The selected AGC assemblies shall be ohm metered at E13 to verify that the DC input resistance of E13 is 150 ohms ± 5 percent. The assemblies shall be tested as follows:

4.7.3.1 Input and output characteristics. Connect test equipment items 1 through 9 as shown in Figure 1. Apply and maintain ± 15.0 Vdc input voltages as specified in Table I, and input video signal amplitude is 0.0 Vdc. Preset R60, R61, R63 and R64 counterclockwise. Preset R62 fully clockwise.

4.7.3.2 AR1 zero. With R60 fully counterclockwise, connect the digital voltmeter to the output of AR1. Short circuit R2 and R3 by attaching jumper wires across these resistors. Meter shall read 0.000 ± .001 Vdc. Remove the jumper wires from across R2 and R3 after test.

4.7.3.3 Prerange gate. Set pulse generator (item 4, Table IV) for a positive pulse of 2.5 V, 10 microseconds (us) wide with a pulse repetition frequency (PRF) of 20 kilohertz (KHz). Using oscilloscope (item 6, Table IV), verify that the prerange gate pulse is present at E6 and E15. With E6 and E15 terminated into 850 ohms, the prerange pulse shall be a negative pulse having a high level voltage of 5.0 ± 1.5 V and a low level voltage of 0.0 ± 0.2 V. Pulse shall be 5.5 ± 2.0 us wide (50% points), and shall have a leading edge fall time of 0.5 ± 0.3 us and a trailing edge rise time of 0.8 ± 0.4 us (10%-90% points).

4.7.3.4 Video gate pulse. Attach a jumper wire from the junction of R19 and C12 to ground. Apply the video input signal and prerange gate trigger specified in Table I by adjusting output amplitude levels and delay on the pulse generators (items 3 and 4, Table IV) for input video signal voltage of +1.5 Vdc. Using oscilloscope (item 6, Table IV) with a 10 megohm probe, observe the pulse at the cathode of CR14. This pulse shall have a positive minimum amplitude of 4.0 V and a negative maximum amplitude of 1.0 V. Pulse rise time shall be 0.5 us maximum. Slope of the pulse top shall not exceed 0.25 V peak-to-peak.

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4.7.3.5 Gain adjustment. With a jumper wire from the junction of R19 and C12 to ground, apply the video input signal and prerange gate trigger specified in Table I. Attach the digital voltmeter (item 5, Table IV) to the output of AR1. Adjust the output amplitude of pulse generator (item 3, Table IV) until the digital voltmeter (item 5, Table IV) reads +1.0 Vdc. Connect the digital voltmeter as specified in Table V (Adjustments). Set each adjustment for the output specified in Table V. (Remove jumper wire after completion of tests.)

TABLE V. Adjustments.

DVM CONNECTION	ADJUSTMENT	OUTPUT
AR2 output (Connect DVM at output of AR3)	R61 (R62 - Full CW)	1.00 ± 0.03 Vdc
AR4 output	R64	1.00 ± 0.03 Vdc

4.7.3.6 AGC tracking. With the inputs to the AGC board assembly as specified in Table I, adjust the amplitude of the video input signal at E14 to obtain a low level voltage of 4.0 ± 0.1 V at E15 as monitored on oscilloscope (item 6, Table IV). Connect oscilloscope to E6 and adjust R62 for a low level voltage of 0.6 ± 0.1 V. When the input signal at E14 is further increased to obtain an output low level at E15 of 6.0 ± 0.1 V, the low level at E6 shall be 2.6 ± 0.2 V.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS10549230.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AGC board is used in the AGC Circuit Assembly (10548228), a subassembly of Radar Receiver R-1475, part of Radar Set AN/VPS-2. The AGC board produces two gated AGC levels. One level may be delayed so that it develops an output only at video inputs above a predetermined amplitude. This delayed AGC level is applied to the IF preamplifier to control its gain.

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6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

AGC circuit board assembly
Fire control assembly
Radar Receiver R-1475
Radar Set AN/VPS-2
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A762)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50679B (AR)	24 August 1990

3. DOCUMENT TITLE
CIRCUIT BOARD ASSEMBLY, AGC: 10549230

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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